



Apparel Assembly / Cut & Sew

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1. Dataset overview

1.1 Scope and activity names

This document describes the life cycle inventory (LCI) model for apparel assembly / cut & sew operations as part of the Carbonfact Textile LCA Database. The published database contains the generic, global (GLO) assembly datasets — one per garment and accessory category. Bangladesh-specific (BD) category datasets were additionally modeled during development and used as a sensitivity check on the generic results; they are not part of the published database.

Bangladesh-specific (BD) datasets — modeled for sensitivity analysis during development, not published:

- assembly tees and tanks
- assembly active tops
- assembly tops, blouses and shirts
- assembly dresses
- assembly sweaters / sweatshirts and fleece
- assembly jeans
- assembly pants
- assembly active pants
- assembly skirts and skorts
- assembly blazers / coats and jackets
- assembly shorts
- assembly one piece
- assembly swim

Published datasets — generic, global (GLO):

- assembly generic t-shirt
- assembly generic shirts

- assembly generic dress
- assembly generic pants
- assembly generic jacket
- assembly generic suit
- assembly generic sweater
- assembly generic swimwear
- assembly generic underwear
- assembly generic bag
- assembly generic accessory
- assembly generic BELT_SHAWL_HAT_BAG_SCARF
- assembly generic houseware
- assembly skirts and skorts
- assembly shorts

1.2 Functional unit and reference flow

Parameter	Value
Functional unit	1 kg of finished garment after assembly / cut & sew, at the mill gate
Reference product	assembly [garment category]
Production amount	1.0
Unit	kilogram
Background database	ecoinvent 3.12, system model: cut-off by classification
Impact method	IPCC 2021 GWP100 via Brightway2 / EF 3.1 midpoint

1.3 Geography

The published datasets use **GLO** (global) geography. The Bangladesh-specific (BD) datasets modeled for sensitivity analysis use **BD** geography, reflecting a major apparel-assembly sourcing region and the origin of the primary energy benchmark literature (Al Mamun et al. 2024).

1.4 Technology representativeness

The datasets represent industrial cut & sew assembly in large-scale garment manufacturing facilities. The energy basis derives from Palamutcu (2010, Turkish textile plants) scaled to garment-making operations, cross-validated with Al Mamun et al. (2024, Bangladesh garment factory). Process steps covered: fabric spreading, cutting, sewing/stitching, ironing/steam pressing, and internal handling.

1.5 Temporal representativeness

Primary energy data: Palamutcu 2010 (Turkish mills) and Al Mamun et al. 2024 (Bangladesh). Auxiliary flow data: Sandin et al. 2019 (Mistra Future Fashion). Cutting waste data: Beton et al. 2014 (IMPRO-Textiles, Table 18). These sources span 2010–2024 and are considered representative of current industrial practice given the stability of cut & sew process parameters over this period.

2. System boundary

2.1 Included foreground operations

- Fabric spreading
- Cutting
- Sewing / stitching operations
- Ironing / steam pressing
- Internal handling directly associated with assembly
- Lubricating oil consumed by assembly machinery
- Sewing thread
- Tissue paper / cutting templates
- Packaging proxy flows (cardboard, LDPE film) where present
- Cutting waste / textile waste treatment
- Waste lubricating oil treatment
- Waste tissue paper treatment
- Water input for steam generation and corresponding wastewater where steam is modelled

2.2 Excluded processes

- Upstream fabric, yarn, and fibre production (modelled in separate spinning, weaving, and knitting datasets)
- Downstream retail, distribution, consumer use, and end-of-life
- Fabric input mass (represented through the IOR/fabric loss relationship — see Section 5.3 — not as a separate material input in the assembly dataset)
- Off-site wastewater treatment infrastructure

2.3 Treatment of fabric input and cutting loss

Fabric input to the assembly stage is not modelled as a technosphere exchange within the assembly dataset itself. The relationship between fabric consumed and garment produced is captured through the **Input–Output Ratio (IOR)**, which scales the upstream fabric production datasets when the assembly module is linked in a full product system. Cutting offcuts are modelled as a waste treatment output within the assembly foreground, using the formula described in Section 5.3. No avoided production credit is applied to cutting waste unless explicitly stated — the current model treats offcuts as burden-bearing waste routed to the ecoinvent waste textile treatment activity.

3. Data sources and literature review

3.1 Source file summary

Source file	Role	Key information extracted
Cut&Sew_DatabaseLCI based on_referencedata.xlsx	Main category-specific foreground inventory	Electricity, steam/heat, water, wastewater, lubricants, sewing thread, tissue paper/templates, cutting waste per garment category

Source file	Role	Key information extracted
Apparel_Assembly_Merged.xlsx	Merged Activity Browser / ecoinvent-ready activity blocks	16 assembly activities with production exchanges, technosphere inputs, waste treatment exchanges
CutSew_Assembly_IOR.xlsx	Input-output ratio and fabric loss table	Waste fractions, IOR (kg fabric/kg garment), loss kg per kg output, source notes
Palamutcu 2010, Energy 35, 2945–2952	Energy benchmark for textile processing stages	Actual and estimated SEC; clothing production SEC 0.065–0.195 kWh/kg
Al Mamun et al. 2024, Manufacturing Letters 41, 1635–1644	Recent garment-making energy model	Real-time energy model for spreading, cutting, sewing and ironing; basic t-shirt total 0.1052 kWh
Sandin & Peters 2018, Journal of Cleaner Production 184, 353–365	Methodological context for textile recycling/reuse and allocation assumptions	Cut-off allocation, waste treatment, inventory transparency

3.2 Palamutcu (2010) — energy benchmark

Palamutcu measures actual and estimated specific electric energy consumption in Turkish cotton textile processing plants using monthly plant records and on-site real-time measurements. The most relevant finding for the present model is that **actual energy consumption per unit output is generally higher than estimated machine-level consumption**, because actual data include inefficiencies, unloaded operation, repair work, product variation, HVAC/lighting interactions and production variability.

Relevant actual SEC values:

Process	Actual SEC
Yarn spinning	3.24–3.47 kWh/kg
Warping/sizing	0.0073 kWh/kg
Weaving	1.58–2.24 kWh/kg
Wet processing	0.79–1.05 kWh/kg
Clothing production	0.065–0.195 kWh/kg

Marginal/additional production electricity requirements:

Process	Value
Yarn production	2.75 MWh/ton
Woven fabric production	1.25 MWh/ton
Wet processing	0.66 MWh/ton
Clothing manufacturing	0.43 MWh/ton

Palamutcu is used primarily as a **benchmark and scaling source** for process-level electricity and heat demand, not as a fully product-specific apparel assembly LCI. The SEC values are Turkey-specific; their use in a Bangladesh context introduces a geographic transfer uncertainty (see Section 12.7).

3.3 Al Mamun et al. (2024) — garment-making energy model

Al Mamun et al. model energy consumption in garment-making operations using real-time data and sewing parameters. The study explicitly critiques SEC and SMV because they include non-productive time and therefore do not accurately represent actual stitching energy. The model links energy consumption to fabric seam length, standard sewing speed, stitching time, machine layout, and motor type.

For a basic knit t-shirt, reported energy consumption:

Sub-process	EC (kWh)
Fabric spreading	0.0038
Fabric cutting	0.0016
Sewing operation	0.0980
Ironing	0.0018
Total garment-making	0.1052

Bangladesh electricity factor used: 570 g CO₂/kWh; resulting

The study also reports that direct-drive servo motors consume substantially less electricity than belt-driven AC systems, and approximately five times less than maximum motor load. Al Mamun et al. is used as a **recent validation source** for the breakdown of garment-making operations. It is Bangladesh-specific and t-shirt-focused — not universally applicable to all garment categories.

3.4 Sandin & Peters (2018) — methodological transparency

Sandin & Peters review textile reuse and recycling LCA studies. The relevant methodological lessons for this assembly model are:

- Cut-off allocation convention: waste treated as burden-free input
- Need for transparent assumptions around waste, system expansion, and replacement rates
- Risk of excluding collection/sorting or other relevant processes
- Inventory transparency as prerequisite for reproducibility

This informs the convention that cutting waste offcuts are treated as burden-bearing waste from the assembly process with no avoided production credit, unless explicitly stated.

3.5 IMPRO-Textiles (Beton et al. 2014) — cutting loss rates

Cutting loss rates by garment category are sourced from IMPRO-Textiles (Beton et al., 2014, JRC EUR 26316 EN, Table 18, original source: ENSAIT). These are listed as spreadsheet-cited references and have not been independently cross-checked against the full report in the present work.

4. Energy model

4.1 Electricity

Electricity consumption per kg finished garment is derived from Palamutcu (2010) category-specific SEC values, expressed in MJ/kg and converted to kWh/kg:

$$\text{Electricity (kWh/kg)} = \frac{\text{SEC (MJ/kg)}}{3.6}$$

Al Mamun et al. (2024) provides independent validation for the t-shirt / active top category (0.1052 kWh for a single garment; consistent with 0.16111 kWh/kg from Palamutcu at typical garment weight).

Electricity is linked to `market group for electricity, medium voltage` for category-specific datasets and at GLO for generic legacy datasets.

4.2 Heat / steam energy

Steam is generated on-site for pressing and ironing operations. Thermal energy is modelled in two steps:

Step 1 — Steam consumption (kg/kg):

$$\text{Steam (kg/kg)} = \frac{\text{Steam energy (MJ/kg)}}{2.26}$$

where 2.26 MJ/kg is the latent heat of vaporisation used throughout this model.

Step 2 — Heat input (MJ/kg) accounting for boiler efficiency:

$$\text{Heat input (MJ/kg)} = \frac{\text{Steam energy (MJ/kg)}}{0.9}$$

Heat is linked to **market group for heat, district or industrial, natural gas** for category-specific datasets.

For categories with no pressing operations (Tees and Tanks, Active Tops, Swim), steam and heat are zero.

4.3 Energy values by garment category

All values are per 1 kg finished garment.

Garment category	Elec (kWh/kg)	MJ_elec source	Steam (kg/kg)	Steam MJ source	Water = Steam (kg/kg)
Tees, Active Tops and Tanks	0.16111	0.58 MJ/kg ÷ 3.6	0	0.0 MJ/kg	0
Tops, Blouses and Shirts	0.31111	1.12 MJ/kg ÷ 3.6	0.10177	0.23 MJ/kg ÷ 2.26	0.10177
Dresses	0.31111	1.12 MJ/kg ÷ 3.6	0.10177	0.23 MJ/kg ÷ 2.26	0.10177
Sweaters, Sweatshirts and Fleece	0.25	0.9 MJ/kg ÷ 3.6	1.24336	2.81 MJ/kg ÷ 2.26	1.24336
Jeans, Pants and Active Pants	0.43889	1.58 MJ/kg ÷ 3.6	0.87611	1.98 MJ/kg ÷ 2.26	0.87611
Skirts and Skorts	0.43889	1.58 MJ/kg ÷ 3.6	0.87611	1.98 MJ/kg ÷ 2.26	0.87611
Blazers, Coats and Jackets	2.01111	7.24 MJ/kg ÷ 3.6	12.48673	28.22 MJ/kg ÷ 2.26	12.48673
Shorts	0.43889	1.58 MJ/kg ÷ 3.6	0.87611	1.98 MJ/kg ÷ 2.26	0.87611
One Piece	0.31111	1.12 MJ/kg ÷ 3.6	0.10177	0.23 MJ/kg ÷ 2.26	0.10177
Swim	0.07222	0.26 MJ/kg ÷ 3.6	0	0.0 MJ/kg	0

5. Inventory construction

5.1 Auxiliary flows — constant across all categories

The following flows are constant across all garment categories (both BD and GLO datasets):

Flow	Amount (kg/kg)	ecoinvent activity	Location	Source
Sewing thread	0.0035	See Flag 5 below	GLO	Sandin et al. 2019 (spreadsheet-cited)
Tissue paper / templates	0.002785	market for tissue paper or market for kraft paper	GLO	Sandin et al. 2019 (spreadsheet-cited)
Lubricating oil	0.000296	lubricating oil production	RoW	Sandin et al. 2019; JRC 2024 (1.44×10 ⁻⁴ kg/kg reference)
Waste mineral oil	0.000296	market for waste mineral oil	RoW	Mirrors lubricating oil input
Waste tissue paper	0.002785	waste paper, unsorted	GLO	Mirrors tissue paper input

5.2 Water and wastewater

Process water is set equal to steam consumption (kg/kg) for categories with pressing operations. For no-steam categories, water and wastewater are zero.

$$\text{Water for steam (kg/kg)} = \text{Steam (kg/kg)}$$

$$\text{Wastewater (m}^3\text{/kg)} = \frac{\text{Water (kg/kg)}}{1000}$$

Water is linked to `market group for tap water {RoW}`. Wastewater is linked to `market for wastewater, average {RoW}`.

5.3 Cutting waste and IOR

Fabric cutting waste is modelled as a waste treatment output per kg finished garment:

$$\text{IOR} = \frac{1}{1 - r}$$

$$\text{Cutting waste (kg/kg output)} = \text{IOR} - 1 = \frac{r}{1 - r}$$

where r is the cutting loss fraction (dimensionless).

Examples:

- 13% loss: $\text{IOR} = 1 / (1 - 0.13) = \mathbf{1.1494}$; waste = **0.1494 kg/kg**
- 18% loss: $\text{IOR} = 1 / (1 - 0.18) = \mathbf{1.2195}$; waste = **0.2195 kg/kg**

Cutting waste is linked to `market for waste yarn and waste textile {RoW}`. No avoided production credit is applied.

5.4 Packaging flows

Cardboard packaging is linked to `market for carton board box production, with gravure printing {GL0}`. LDPE packaging film (`market for packaging film, low density polyethylene {GL0}`) is present in merged activity blocks but set to zero in most activities.

6. Calculation rules

All conversion rules applied consistently throughout the model:

$$\text{Electricity (kWh/kg)} = \frac{\text{Electricity (MJ/kg)}}{3.6}$$

$$\text{Steam (kg/kg)} = \frac{\text{Steam energy (MJ/kg)}}{2.26}$$

where 2.26 MJ/kg is the latent heat of vaporisation at standard conditions.

$$\text{Water for steam (kg/kg)} = \text{Steam (kg/kg)}$$

$$\text{Wastewater (m}^3\text{/kg)} = \frac{\text{Water (kg/kg)}}{1000}$$

$$\text{IOR} = \frac{1}{1 - r} \quad \text{Cutting waste (kg/kg)} = \frac{r}{1 - r}$$

$$\text{Heat input to boiler (MJ/kg)} = \frac{\text{Steam energy (MJ/kg)}}{0.9}$$

7. Input-Output Ratio (IOR)

$$\text{IOR} = \frac{1}{1 - r}$$

Table 7. Fabric IOR by garment category

Process / Garment Category	Location	Waste (%)	IOR (kg fabric/kg garment)	Input required (kg)	Loss (kg/kg output)	Source
Generic T-shirt (assembly)	GLO	13%	1.1494	1.1494	0.1494	Sandin et al. 2019; Baseimpact energy data
Generic Pants (assembly)	GLO	14%	1.1628	1.1628	0.1628	Sandin et al. 2019; Baseimpact energy data
Generic Shirt (assembly)	GLO	13%	1.1494	1.1494	0.1494	Sandin et al. 2019; Baseimpact energy data
Generic Jacket (assembly)	GLO	16%	1.1905	1.1905	0.1905	Sandin et al. 2019; Baseimpact energy data
Generic Dress (assembly)	GLO	18%	1.2195	1.2195	0.2195	Sandin et al. 2019; Baseimpact energy data
Generic Sweater (assembly)	GLO	10%	1.1111	1.1111	0.1111	Sandin et al. 2019; Baseimpact energy data
Generic Swimwear (assembly)	GLO	18%	1.2195	1.2195	0.2195	Sandin et al. 2019; Baseimpact energy data
Generic Underwear (assembly)	GLO	13%	1.1494	1.1494	0.1494	Sandin et al. 2019; Baseimpact energy data
Generic Suit (assembly)	GLO	16%	1.1905	1.1905	0.1905	Sandin et al. 2019; Baseimpact energy data
Generic Bag (assembly)	GLO	10%	1.1111	1.1111	0.1111	Sandin et al. 2019; Baseimpact energy data
Generic Belt/Shawl/Hat/Scarf (assembly)	GLO	10%	1.1111	1.1111	0.1111	Sandin et al. 2019; Baseimpact energy data
Generic Accessory (assembly)	GLO	10%	1.1111	1.1111	0.1111	Sandin et al. 2019; Baseimpact / PEFCR RP10

Process / Garment Category	Location	Waste (%)	IOR (kg fabric/kg garment)	Input required (kg)	Loss (kg/kg output)	Source
Skirts and Skorts (assembly)	GLO	14%	1.1628	1.1628	0.1628	Sandin et al. 2019; Baseimpact energy data
Shorts (assembly)	GLO	15%	1.1765	1.1765	0.1765	Sandin et al. 2019; Baseimpact energy data
Houseware (assembly)	GLO	16.7%	1.2000	1.2000	0.2000	Baseimpact / PEFCR RP10

8. Category-specific inventories (BD datasets)

All values are per 1 kg finished garment. Constant auxiliary flows (thread 0.0035 kg, tissue 0.002785 kg, lubricant 0.000296 kg, waste oil 0.000296 kg, waste tissue 0.002785 kg) are omitted from the energy/water summary but included in the full foreground exchange tables in Section 9.

Table 8. Energy, water, and cutting waste summary — category-specific BD datasets

Category	Elec (kWh/kg)	Steam (kWh/kg)	Water (kg/kg)	Waste (%)	Cutting waste (kg/kg)
Tees, Active Tops and Tanks	0.16111	0	0	13%	0.14943
Tops, Blouses and Shirts	0.31111	0.10177	0.10177	13%	0.14943
Dresses	0.31111	0.10177	0.10177	18%	0.21951
Sweaters, Sweatshirts and Fleece	0.25	1.24336	1.24336	10%	0.11111
Jeans, Pants and Active Pants	0.43889	0.87611	0.87611	14%	0.16279
Skirts and Skorts	0.43889	0.87611	0.87611	14%	0.16279
Blazers, Coats and Jackets	2.01111	12.48673	12.48673	16%	0.19048
Shorts	0.43889	0.87611	0.87611	15%	0.17647
One Piece	0.31111	0.10177	0.10177	14%	0.16279
Swim	0.07222	0	0	18%	0.21951

9. Full foreground exchange tables

9.1 DB_Cut&Sew — one exchange per row (all categories)

Garment category	Flow group	Amount	Unit	Activity / proxy	Location
Tees, Active Tops and Tanks	Electricity	0.16111	kWh	market group for electricity, medium voltage	BD
Tees, Active Tops and Tanks	Heat/Steam energy	0	kWh	market group for heat, district or industrial, natural gas	BD
Tees, Active Tops and Tanks	Lubricating oil	0.000296	kg	lubricating oil production	RoW

Garment category	Flow group	Amount	Unit	Activity / proxy	Location
Tees, Active Tops and Tanks	Waste mineral oil	0.000296	kg	market for waste mineral oil	RoW
Tees, Active Tops and Tanks	Cutting/textile waste	0.14943	kg	market for waste yarn and waste textile	RoW
Tees, Active Tops and Tanks	Thread	0.0035	kg	market for yarn, cotton [see Flag 5]	GLO
Tees, Active Tops and Tanks	Tissue paper	0.002785	kg	market for kraft paper	GLO
Tees, Active Tops and Tanks	Waste tissue	0.002785	kg	waste paper, unsorted	GLO
Tees, Active Tops and Tanks	Water	0	kg	market group for tap water	RoW
Tees, Active Tops and Tanks	Wastewater	0	m ³	market for wastewater, average	RoW
Tops, Blouses and Shirts	Electricity	0.31111	kWh	market group for electricity, medium voltage	BD
Tops, Blouses and Shirts	Heat/Steam energy	0.10177	kWh	market group for heat, district or industrial, natural gas	BD
Tops, Blouses and Shirts	Lubricating oil	0.000296	kg	lubricating oil production	RoW
Tops, Blouses and Shirts	Waste mineral oil	0.000296	kg	market for waste mineral oil	RoW
Tops, Blouses and Shirts	Cutting/textile waste	0.14943	kg	market for waste yarn and waste textile	RoW
Tops, Blouses and Shirts	Thread	0.0035	kg	market for yarn, cotton [see Flag 5]	GLO
Tops, Blouses and Shirts	Tissue paper	0.002785	kg	market for kraft paper	GLO
Tops, Blouses and Shirts	Waste tissue	0.002785	kg	waste paper, unsorted	GLO
Tops, Blouses and Shirts	Water	0.10177	kg	market group for tap water	RoW
Tops, Blouses and Shirts	Wastewater	0.00010177	m ³	market for wastewater, average	RoW
Dresses	Electricity	0.31111	kWh	market group for electricity, medium voltage	BD
Dresses	Heat/Steam energy	0.10177	kWh	market group for heat, district or industrial, natural gas	BD
Dresses	Lubricating oil	0.000296	kg	lubricating oil production	RoW
Dresses	Waste mineral oil	0.000296	kg	market for waste mineral oil	RoW
Dresses	Cutting/textile waste	0.21951	kg	market for waste yarn and waste textile	RoW
Dresses	Thread	0.0035	kg	market for yarn, cotton [see Flag 5]	GLO
Dresses	Tissue paper	0.002785	kg	market for kraft paper	GLO
Dresses	Waste tissue	0.002785	kg	waste paper, unsorted	GLO

Garment category	Flow group	Amount	Unit	Activity / proxy	Location
Dresses	Water	0.10177	kg	market group for tap water	RoW
Dresses	Wastewater	0.00010177	m³	market for wastewater, average	RoW
Sweaters, Sweatshirts and Fleece	Electricity	0.25	kWh	market group for electricity, medium voltage	BD
Sweaters, Sweatshirts and Fleece	Heat/Steam energy	1.24336	kWh	market group for heat, district or industrial, natural gas	BD
Sweaters, Sweatshirts and Fleece	Lubricating oil	0.000296	kg	lubricating oil production	RoW
Sweaters, Sweatshirts and Fleece	Waste mineral oil	0.000296	kg	market for waste mineral oil	RoW
Sweaters, Sweatshirts and Fleece	Cutting/textile waste	0.1111	kg	market for waste yarn and waste textile	RoW
Sweaters, Sweatshirts and Fleece	Thread	0.0035	kg	market for yarn, cotton [see Flag 5]	GLO
Sweaters, Sweatshirts and Fleece	Tissue paper	0.002785	kg	market for kraft paper	GLO
Sweaters, Sweatshirts and Fleece	Waste tissue	0.002785	kg	waste paper, unsorted	GLO
Sweaters, Sweatshirts and Fleece	Water	1.24336	kg	market group for tap water	RoW
Sweaters, Sweatshirts and Fleece	Wastewater	0.00124336	m³	market for wastewater, average	RoW
Jeans, Pants and Active Pants	Electricity	0.43889	kWh	market group for electricity, medium voltage	BD
Jeans, Pants and Active Pants	Heat/Steam energy	0.87611	kWh	market group for heat, district or industrial, natural gas	BD
Jeans, Pants and Active Pants	Lubricating oil	0.000296	kg	lubricating oil production	RoW
Jeans, Pants and Active Pants	Waste mineral oil	0.000296	kg	market for waste mineral oil	RoW
Jeans, Pants and Active Pants	Cutting/textile waste	0.16279	kg	market for waste yarn and waste textile	RoW
Jeans, Pants and Active Pants	Thread	0.0035	kg	market for yarn, cotton [see Flag 5]	GLO
Jeans, Pants and Active Pants	Tissue paper	0.002785	kg	market for kraft paper	GLO
Jeans, Pants and Active Pants	Waste tissue	0.002785	kg	waste paper, unsorted	GLO
Jeans, Pants and Active Pants	Water	0.87611	kg	market group for tap water	RoW
Jeans, Pants and Active Pants	Wastewater	0.0087611	m³	market for wastewater, average	RoW
Skirts and Skorts	Electricity	0.43889	kWh	market group for electricity, medium voltage	BD
Skirts and Skorts	Heat/Steam energy	0.87611	kWh	market group for heat, district or industrial, natural gas	BD

Garment category	Flow group	Amount	Unit	Activity / proxy	Location
Skirts and Skorts	Lubricating oil	0.000296	kg	lubricating oil production	RoW
Skirts and Skorts	Waste mineral oil	0.000296	kg	market for waste mineral oil	RoW
Skirts and Skorts	Cutting/textile waste	0.16279	kg	market for waste yarn and waste textile	RoW
Skirts and Skorts	Thread	0.0035	kg	market for yarn, cotton [see Flag 5]	GLO
Skirts and Skorts	Tissue paper	0.002785	kg	market for kraft paper	GLO
Skirts and Skorts	Waste tissue	0.002785	kg	waste paper, unsorted	GLO
Skirts and Skorts	Water	0.87611	kg	market group for tap water	RoW
Skirts and Skorts	Wastewater	0.0087611	m³	market for wastewater, average	RoW
Blazers, Coats and Jackets	Electricity	2.01111	kWh	market group for electricity, medium voltage	BD
Blazers, Coats and Jackets	Heat/Steam energy	12.48673	kWh	market group for heat, district or industrial, natural gas	BD
Blazers, Coats and Jackets	Lubricating oil	0.000296	kg	lubricating oil production	RoW
Blazers, Coats and Jackets	Waste mineral oil	0.000296	kg	market for waste mineral oil	RoW
Blazers, Coats and Jackets	Cutting/textile waste	0.19048	kg	market for waste yarn and waste textile	RoW
Blazers, Coats and Jackets	Thread	0.0035	kg	market for yarn, cotton [see Flag 5]	GLO
Blazers, Coats and Jackets	Tissue paper	0.002785	kg	market for kraft paper	GLO
Blazers, Coats and Jackets	Waste tissue	0.002785	kg	waste paper, unsorted	GLO
Blazers, Coats and Jackets	Water	12.48673	kg	market group for tap water	RoW
Blazers, Coats and Jackets	Wastewater	0.01248673	m³	market for wastewater, average	RoW
Shorts	Electricity	0.43889	kWh	market group for electricity, medium voltage	BD
Shorts	Heat/Steam energy	0.87611	kWh	market group for heat, district or industrial, natural gas	BD
Shorts	Lubricating oil	0.000296	kg	lubricating oil production	RoW
Shorts	Waste mineral oil	0.000296	kg	market for waste mineral oil	RoW
Shorts	Cutting/textile waste	0.17647	kg	market for waste yarn and waste textile	RoW
Shorts	Thread	0.0035	kg	market for yarn, cotton [see Flag 5]	GLO
Shorts	Tissue paper	0.002785	kg	market for kraft paper	GLO
Shorts	Waste tissue	0.002785	kg	waste paper, unsorted	GLO
Shorts	Water	0.87611	kg	market group for tap water	RoW

Garment category	Flow group	Amount	Unit	Activity / proxy	Location
Shorts	Wastewater	0.0087611	m ³	market for wastewater, average	RoW
One Piece	Electricity	0.31111	kWh	market group for electricity, medium voltage	BD
One Piece	Heat/Steam energy	0.10177	kWh	market group for heat, district or industrial, natural gas	BD
One Piece	Lubricating oil	0.000296	kg	lubricating oil production	RoW
One Piece	Waste mineral oil	0.000296	kg	market for waste mineral oil	RoW
One Piece	Cutting/textile waste	0.16279	kg	market for waste yarn and waste textile	RoW
One Piece	Thread	0.0035	kg	market for yarn, cotton [see Flag 5]	GLO
One Piece	Tissue paper	0.002785	kg	market for kraft paper	GLO
One Piece	Waste tissue	0.002785	kg	waste paper, unsorted	GLO
One Piece	Water	0.10177	kg	market group for tap water	RoW
One Piece	Wastewater	0.0010177	m ³	market for wastewater, average	RoW
Swim	Electricity	0.07222	kWh	market group for electricity, medium voltage	BD
Swim	Heat/Steam energy	0	kWh	market group for heat, district or industrial, natural gas	BD
Swim	Lubricating oil	0.000296	kg	lubricating oil production	RoW
Swim	Waste mineral oil	0.000296	kg	market for waste mineral oil	RoW
Swim	Cutting/textile waste	0.21951	kg	market for waste yarn and waste textile	RoW
Swim	Thread	0.0035	kg	market for yarn, cotton [see Flag 5]	GLO
Swim	Tissue paper	0.002785	kg	market for kraft paper	GLO
Swim	Waste tissue	0.002785	kg	waste paper, unsorted	GLO
Swim	Water	0	kg	market group for tap water	RoW
Swim	Wastewater	0	m ³	market for wastewater, average	RoW

9.2 Apparel_Assembly_Merged — full activity blocks (GLO datasets)

assembly generic t-shirt | BD | 1 kg finished garment

Name	Amount	Unit	Location	Type	Comment
assembly generic t-shirt	1	kg	BD	production	—
market group for electricity, medium voltage	0.16111	kWh	BD	technosphere	Cut&Sew — Tees, Active Tops and Tanks. Source: Palamutcu 2010; Al Mamun et al. 2024.
market group for heat, district or industrial, natural gas	0	kWh	BD	technosphere	Cut&Sew — Tees, Active Tops and Tanks. Steam energy. Source: Palamutcu 2010.

Name	Amount	Unit	Location	Type	Comment
market for yarn, cotton	0.0035	kg	GLO	technosphere	Sewing threads — Sandin et al. 2019
market for tissue paper	0.002785	kg	GLO	technosphere	Confectioning template — Sandin et al. 2019
market for carton board box production, with gravure printing	0.000703	kg	GLO	technosphere	Corrugated board box — Sandin et al. 2019 [see Flag 3]
market for packaging film, low density polyethylene	0	kg	GLO	technosphere	Packaging film — Sandin et al. 2020
lubricating oil production	0.000296	kg	RoW	technosphere	Lubricant consumption during assembly — Sandin et al. 2019
market group for tap water	0	kg	RoW	technosphere	Cut&Sew — water = steam equivalent. Reviewed: original 24.72 kg/kg flagged as too high.
market for wastewater, average	0	m ³	RoW	technosphere	—
waste paper, unsorted	0.002785	kg	GLO	technosphere	Solid waste — Tissue paper waste — Sandin et al. 2019
market for waste mineral oil	0.000296	kg	RoW	technosphere	Solid waste — Lubricant waste — Sandin et al. 2019
market for waste yarn and waste textile	0.14943	kg	RoW	technosphere	Cutting waste — Tees, Active Tops and Tanks. Rate=13.0%. Source: IMPRO-Textiles (Beton et al. 2014).

assembly generic dress | BD | 1 kg finished garment

Name	Amount	Unit	Location	Type	Comment
assembly generic dress	1	kg	BD	production	—
market group for electricity, medium voltage	0.31111	kWh	BD	technosphere	Cut&Sew — Dresses. Source: Palamutcu 2010; Al Mamun et al. 2024.
market group for heat, district or industrial, natural gas	0.10177	kWh	BD	technosphere	Cut&Sew — Dresses. Steam energy. Source: Palamutcu 2010.
market for yarn, cotton	0.0035	kg	GLO	technosphere	Sewing threads — Sandin et al. 2019
market for tissue paper	0.002785	kg	GLO	technosphere	Confectioning template — Sandin et al. 2019
market for carton board box production, with gravure printing	0.000703	kg	GLO	technosphere	Corrugated board box — Sandin et al. 2019 [see Flag 3]
market for packaging film, low density polyethylene	0	kg	GLO	technosphere	—
lubricating oil production	0.000296	kg	RoW	technosphere	Lubricant consumption — Sandin et al. 2019

Name	Amount	Unit	Location	Type	Comment
market group for tap water	0.10177	kg	RoW	technosphere	water = steam equivalent
market for wastewater, average	0.00010177	m ³	RoW	technosphere	—
waste paper, unsorted	0.002785	kg	GLO	technosphere	Tissue paper waste — Sandin et al. 2019
market for waste mineral oil	0.000296	kg	RoW	technosphere	Lubricant waste — Sandin et al. 2019
market for waste yarn and waste textile	0.21951	kg	RoW	technosphere	Cutting waste — Dresses. Rate=18.0%. Source: IMPRO-Textiles (Beton et al. 2014).

assembly generic pants | BD | 1 kg finished garment

Name	Amount	Unit	Location	Type	Comment
assembly generic pants	1	kg	BD	production	—
market group for electricity, medium voltage	0.43889	kWh	BD	technosphere	Cut&Sew — Jeans, Pants and Active Pants. Source: Palamutcu 2010; Al Mamun et al. 2024.
market group for heat, district or industrial, natural gas	0.87611	kWh	BD	technosphere	Cut&Sew — Jeans, Pants and Active Pants. Steam energy. Source: Palamutcu 2010.
market for yarn, cotton	0.0035	kg	GLO	technosphere	Sewing threads — Sandin et al. 2019
market for tissue paper	0.002785	kg	GLO	technosphere	Confectioning template — Sandin et al. 2019
market for carton board box production, with gravure printing	0.000703	kg	GLO	technosphere	[see Flag 3]
market for packaging film, low density polyethylene	0	kg	GLO	technosphere	—
lubricating oil production	0.000296	kg	RoW	technosphere	Lubricant consumption — Sandin et al. 2019
market group for tap water	0.87611	kg	RoW	technosphere	water = steam equivalent
market for wastewater, average	0.0087611	m ³	RoW	technosphere	—
waste paper, unsorted	0.002785	kg	GLO	technosphere	Tissue paper waste
market for waste mineral oil	0.000296	kg	RoW	technosphere	Lubricant waste
market for waste yarn and waste textile	0.16279	kg	RoW	technosphere	Cutting waste — Rate=14.0%. Source: IMPRO-Textiles (Beton et al. 2014).

assembly generic jacket | BD | 1 kg finished garment

Name	Amount	Unit	Location	Type	Comment
assembly generic jacket	1	kg	BD	production	—
market group for electricity, medium voltage	2.01111	kWh	BD	technosphere	Cut&Sew — Blazers, Coats and Jackets. Source: Palamutcu 2010; Al Mamun et al. 2024.
market group for heat, district or industrial, natural gas	12.48673	kWh	BD	technosphere	Cut&Sew — Blazers, Coats and Jackets. Steam energy. Source: Palamutcu 2010.
market for yarn, cotton	0.0035	kg	GLO	technosphere	Sewing threads — Sandin et al. 2019
market for tissue paper	0.002785	kg	GLO	technosphere	Confectioning template — Sandin et al. 2019
market for carton board box production, with gravure printing	0.000703	kg	GLO	technosphere	[see Flag 3]
market for packaging film, low density polyethylene	0	kg	GLO	technosphere	—
lubricating oil production	0.000296	kg	RoW	technosphere	Lubricant consumption — Sandin et al. 2019
market group for tap water	12.48673	kg	RoW	technosphere	water = steam equivalent
market for wastewater, average	0.01248673	m ³	RoW	technosphere	—
waste paper, unsorted	0.002785	kg	GLO	technosphere	Tissue paper waste
market for waste mineral oil	0.000296	kg	RoW	technosphere	Lubricant waste
market for waste yarn and waste textile	0.19048	kg	RoW	technosphere	Cutting waste — Rate=16.0%. Source: IMPRO-Textiles (Beton et al. 2014).

assembly generic sweater | BD | 1 kg finished garment

Name	Amount	Unit	Location	Type	Comment
assembly generic sweater	1	kg	BD	production	—
market group for electricity, medium voltage	0.25	kWh	BD	technosphere	Cut&Sew — Sweaters. Source: Palamutcu 2010; Al Mamun et al. 2024.
market group for heat, district or industrial, natural gas	1.24336	kWh	BD	technosphere	Cut&Sew — Sweaters. Steam energy. Source: Palamutcu 2010.
market for yarn, cotton	0.0035	kg	GLO	technosphere	Sewing threads
market for tissue paper	0.002785	kg	GLO	technosphere	Confectioning template
market for carton board box production, with gravure printing	0.000703	kg	GLO	technosphere	[see Flag 3]
market for packaging film, low density polyethylene	0	kg	GLO	technosphere	—

Name	Amount	Unit	Location	Type	Comment
lubricating oil production	0.000296	kg	RoW	technosphere	Lubricant consumption
market group for tap water	1.24336	kg	RoW	technosphere	water = steam equivalent
market for wastewater, average	0.00124336	m ³	RoW	technosphere	—
waste paper, unsorted	0.002785	kg	GLO	technosphere	Tissue paper waste
market for waste mineral oil	0.000296	kg	RoW	technosphere	Lubricant waste
market for waste yarn and waste textile	0.11111	kg	RoW	technosphere	Cutting waste — Rate=10.0%. Source: IMPRO-Textiles (Beton et al. 2014).

assembly generic swimwear | BD | 1 kg finished garment

Name	Amount	Unit	Location	Type	Comment
assembly generic swimwear	1	kg	BD	production	—
market group for electricity, medium voltage	0.07222	kWh	BD	technosphere	Cut&Sew — Swim. Source: Palamutcu 2010; Al Mamun et al. 2024.
market group for heat, district or industrial, natural gas	0	kWh	BD	technosphere	No steam — Swim. Source: Palamutcu 2010.
market for yarn, cotton	0.0035	kg	GLO	technosphere	Sewing threads
market for tissue paper	0.002785	kg	GLO	technosphere	Confectioning template
market for carton board box production, with gravure printing	0.000703	kg	GLO	technosphere	[see Flag 3]
market for packaging film, low density polyethylene	0	kg	GLO	technosphere	—
lubricating oil production	0.000296	kg	RoW	technosphere	Lubricant consumption
market group for tap water	0	kg	RoW	technosphere	No steam, no water
market for wastewater, average	0	m ³	RoW	technosphere	—
waste paper, unsorted	0.002785	kg	GLO	technosphere	Tissue paper waste
market for waste mineral oil	0.000296	kg	RoW	technosphere	Lubricant waste
market for waste yarn and waste textile	0.21951	kg	RoW	technosphere	Cutting waste — Rate=18.0%. Source: IMPRO-Textiles (Beton et al. 2014).

assembly generic shirts | BD | 1 kg finished garment

Name	Amount	Unit	Location	Type	Comment
assembly generic shirts	1	kg	BD	production	—

Name	Amount	Unit	Location	Type	Comment
market group for electricity, medium voltage	0.31111	kWh	BD	technosphere	Cut&Sew — Tops, Blouses and Shirts. Source: Palamutcu 2010; Al Mamun et al. 2024.
market group for heat, district or industrial, natural gas	0.10177	kWh	BD	technosphere	Steam energy. Source: Palamutcu 2010.
market for yarn, cotton	0.0035	kg	GLO	technosphere	Sewing threads
market for tissue paper	0.002785	kg	GLO	technosphere	Confectioning template
market for carton board box production, with gravure printing	0.000703	kg	GLO	technosphere	[see Flag 3]
market for packaging film, low density polyethylene	0	kg	GLO	technosphere	—
lubricating oil production	0.000296	kg	RoW	technosphere	Lubricant consumption
market group for tap water	0.10177	kg	RoW	technosphere	water = steam equivalent
market for wastewater, average	0.00010177	m³	RoW	technosphere	—
waste paper, unsorted	0.002785	kg	GLO	technosphere	Tissue paper waste
market for waste mineral oil	0.000296	kg	RoW	technosphere	Lubricant waste
market for waste yarn and waste textile	0.14943	kg	RoW	technosphere	Cutting waste — Rate=13.0%. Source: IMPRO-Textiles (Beton et al. 2014).

assembly generic suit | BD | 1 kg finished garment

Name	Amount	Unit	Location	Type	Comment
assembly generic suit	1	kg	BD	production	—
market group for electricity, medium voltage	2.01111	kWh	BD	technosphere	Cut&Sew — Blazers, Coats and Jackets. Source: Palamutcu 2010; Al Mamun et al. 2024.
market group for heat, district or industrial, natural gas	12.48673	kWh	BD	technosphere	Steam energy. Source: Palamutcu 2010.
market for yarn, cotton	0.0035	kg	GLO	technosphere	Sewing threads
market for tissue paper	0.002785	kg	GLO	technosphere	Confectioning template
market for carton board box production, with gravure printing	0.000703	kg	GLO	technosphere	[see Flag 3]
market for packaging film, low density polyethylene	0	kg	GLO	technosphere	—
lubricating oil production	0.000296	kg	RoW	technosphere	Lubricant consumption
market group for tap water	12.48673	kg	RoW	technosphere	water = steam equivalent

Name	Amount	Unit	Location	Type	Comment
market for wastewater, average	0.01248673	m³	RoW	technosphere	—
waste paper, unsorted	0.002785	kg	GLO	technosphere	Tissue paper waste
market for waste mineral oil	0.000296	kg	RoW	technosphere	Lubricant waste
market for waste yarn and waste textile	0.19048	kg	RoW	technosphere	Cutting waste — Rate=16.0%. Source: IMPRO-Textiles (Beton et al. 2014).

assembly generic underwear | GLO | 1 kg finished garment

Name	Amount	Unit	Location	Type	Comment
assembly generic underwear	1	kg	GLO	production	—
heat mix	0	MJ	GLO	technosphere	base impact
market group for electricity, medium voltage	6.25	kWh	GLO	technosphere	base impact, adjusted Underwear (RP8): 80 g/product from PEFCR
market for yarn, cotton	0.0035	kg	GLO	technosphere	Sewing threads — Sandin et al. 2019
market for tissue paper	0.002785	kg	GLO	technosphere	Confectioning template
market for carton board box production, with gravure printing	0.000703	kg	GLO	technosphere	[see Flag 3]
market for packaging film, low density polyethylene	0	kg	GLO	technosphere	—
market for lubricating oil	0.000296	kg	RoW	technosphere	Lubricant consumption
market for tap water	24.72	kg	RoW	technosphere	[FLAG — original supplier value; to be updated]
market for wastewater from textile production	0.02472	m³	GLO	technosphere	—
market for waste packaging paper	0.003488	kg	RoW	technosphere	Carton waste from packaging
treatment of waste polyethylene terephthalate, municipal incineration	0	kg	GLO	technosphere	Waste plastic packaging

assembly generic accessory | GLO | 1 kg finished garment

Name	Amount	Unit	Location	Type	Comment
assembly generic accessory	1	kg	GLO	production	—
heat mix	0	MJ	GLO	technosphere	base impact
market group for electricity, medium voltage	4.5455	kWh	GLO	technosphere	base impact, adjusted apparel accessories (RP10): 110 g/product from PEFCR
market for yarn, cotton	0.0035	kg	GLO	technosphere	Sewing threads — Sandin et al. 2019

Name	Amount	Unit	Location	Type	Comment
market for tissue paper	0.002785	kg	GLO	technosphere	Confectioning template
market for carton board box production, with gravure printing	0.000703	kg	GLO	technosphere	[see Flag 3]
market for packaging film, low density polyethylene	0	kg	GLO	technosphere	—
market for lubricating oil	0.000296	kg	RoW	technosphere	Lubricant consumption
market for tap water	24.72	kg	RoW	technosphere	[FLAG — original supplier value; to be updated]
market for wastewater from textile production	0.02472	m³	GLO	technosphere	—
market for waste packaging paper	0.003488	kg	RoW	technosphere	Carton waste from packaging
treatment of waste polyethylene terephthalate, municipal incineration	0	kg	GLO	technosphere	Waste plastic packaging

assembly generic bag | GLO | 1 kg finished garment

Name	Amount	Unit	Location	Type	Comment
assembly generic bag	1	kg	GLO	production	—
heat mix	0	MJ	GLO	technosphere	base impact
market group for electricity, medium voltage	3.164	kWh	GLO	technosphere	base impact
market for yarn, cotton	0.0035	kg	GLO	technosphere	Sewing threads
market for tissue paper	0.002785	kg	GLO	technosphere	Confectioning template
market for carton board box production, with gravure printing	0.000703	kg	GLO	technosphere	[see Flag 3]
market for packaging film, low density polyethylene	0	kg	GLO	technosphere	—
market for lubricating oil	0.000296	kg	RoW	technosphere	Lubricant consumption
market for tap water	24.72	kg	RoW	technosphere	[FLAG — original supplier value; to be updated]
market for wastewater from textile production	0.02472	m³	GLO	technosphere	—
market for waste packaging paper	0.003488	kg	RoW	technosphere	Carton waste from packaging
treatment of waste polyethylene terephthalate, municipal incineration	0	kg	GLO	technosphere	Waste plastic packaging

assembly generic BELT_SHAWL_HAT_BAG_SCARF | GLO | 1 kg finished garment

Name	Amount	Unit	Location	Type	Comment
assembly generic BELT_SHAWL_HAT_BAG_SCARF	1	kg	GLO	production	—
heat mix	0	MJ	GLO	technosphere	base impact
market group for electricity, medium voltage	3.8127	kWh	GLO	technosphere	base impact, adjusted apparel accessories (RP10): 110 g/product from PEFCR
market for yarn, cotton	0.0035	kg	GLO	technosphere	Sewing threads
market for tissue paper	0.002785	kg	GLO	technosphere	Confectioning template
market for carton board box production, with gravure printing	0.000703	kg	GLO	technosphere	[see Flag 3]
market for packaging film, low density polyethylene	0	kg	GLO	technosphere	—
market for lubricating oil	0.000296	kg	RoW	technosphere	Lubricant consumption
market for tap water	24.72	kg	RoW	technosphere	[FLAG — original supplier value; to be updated]
market for wastewater from textile production	0.02472	m ³	GLO	technosphere	—
market for waste packaging paper	0.003488	kg	RoW	technosphere	Carton waste from packaging
treatment of waste polyethylene terephthalate, municipal incineration	0	kg	GLO	technosphere	Waste plastic packaging

assembly generic houseware | GLO | 1 kg finished garment

Name	Amount	Unit	Location	Type	Comment
assembly generic houseware	1	kg	GLO	production	—
heat mix	0	MJ	GLO	technosphere	base impact
market group for electricity, medium voltage	9.0909	kWh	GLO	technosphere	base impact, adjusted apparel accessories (RP10): 110 g/product from PEFCR
market for yarn, cotton	0.0035	kg	GLO	technosphere	Sewing threads
market for tissue paper	0.002785	kg	GLO	technosphere	Confectioning template
market for carton board box production, with gravure printing	0.000703	kg	GLO	technosphere	[see Flag 3]
market for packaging film, low density polyethylene	0	kg	GLO	technosphere	—
market for lubricating oil	0.000296	kg	RoW	technosphere	Lubricant consumption
market for tap water	24.72	kg	RoW	technosphere	[FLAG — original supplier value; to be updated]

Name	Amount	Unit	Location	Type	Comment
market for wastewater from textile production	0.02472	m ³	GLO	technosphere	—
market for waste packaging paper	0.003488	kg	RoW	technosphere	Carton waste from packaging
treatment of waste polyethylene terephthalate, municipal incineration	0	kg	GLO	technosphere	Waste plastic packaging

10. Proxy choices and flow conventions

Flow	Preferred treatment	ecoinvent activity	Notes
Electricity	Technosphere input per kg	market group for electricity, medium voltage	BD for category-specific; GLO for generic legacy
Heat / steam	Technosphere input per kg	market group for heat, district or industrial, natural gas	[Flag 2] Units: kWh in database; verify against steam MJ derivation
Water for steam	Technosphere water input	market group for tap water	Amount = steam kg/kg; zero for no-steam categories
Wastewater	Waste treatment output	market for wastewater, average	Amount in m ³ /kg = water kg/kg ÷ 1000
Lubricating oil	Technosphere input	lubricating oil production {RoW}	Amount: 0.000296 kg/kg. Note: legacy GLO uses market for lubricating oil
Waste mineral oil	Waste treatment output	market for waste mineral oil {RoW}	Amount mirrors lubricating oil input
Sewing thread	Technosphere input	market for yarn, cotton {GLO}	[Flag 5] Check consistency with polyester fibre waste proxy in some files
Tissue paper/templates	Technosphere input	market for tissue paper {GLO} or market for kraft paper {GLO}	Amount: 0.002785 kg/kg
Waste tissue	Waste treatment output	waste paper, unsorted {GLO}	Amount mirrors tissue paper input
Cutting/textile waste	Waste treatment output	market for waste yarn and waste textile {RoW}	Amount = $r / (1 - r)$ per kg output
Cardboard packaging	Technosphere input	market for carton board box production, with gravure printing {GLO}	[Flag 3] Amount: 0.000703 kg/kg in merged blocks vs. 1.0 kg/kg in analysis notes — unresolved
LDPE packaging film	Technosphere input	market for packaging film, low density polyethylene {GLO}	Set to 0 in most activities

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